



BAHRIA UNIVERSITY ISLAMABAD

Department of Computer Science

DESIGN and ANALYSIS of ALGORITHMS

ASSIGNMENT # 4

Submitted By

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Project Part II

Celebrity Puzzle Game:

1. Problem Statement:

The Celebrity Problem is a logical puzzle in which a celebrity is a person who is known by everyone in the group but does not know anyone else. The task is to identify the celebrity by asking questions in the form of:

“Does person A know person B?”

The game allows the user to ask questions, analyze the relationship between people and guess the celebrity by using logical reasoning and algorithm techniques.

2. Objective:

- The main objectives of this project are:
- To develop an interactive game based on the Celebrity Puzzle
- To implement the algorithms designed in part I of the project
- To provide a user-friendly interface
- To improve understanding of algorithm design techniques

3. Tools & Language Used:

Programming Language

- Python

GUI Library

- Tkinter

4. Game Working:

The game starts by generating a group of people where one hidden person is selected as the celebrity. The user interacts with the game through graphical interface. The player can ask the question like: “Does Person A know Person B?”

The system answers with YES or NO based on the hidden relationship matrix.

The user analyzes the answers and tries to identify the celebrity. The game also provides options to run Brute force algorithm and the Decrease-and-Conquer algorithm to compare their performances. A score system is included, where a few questions give a higher score.

5. Code:

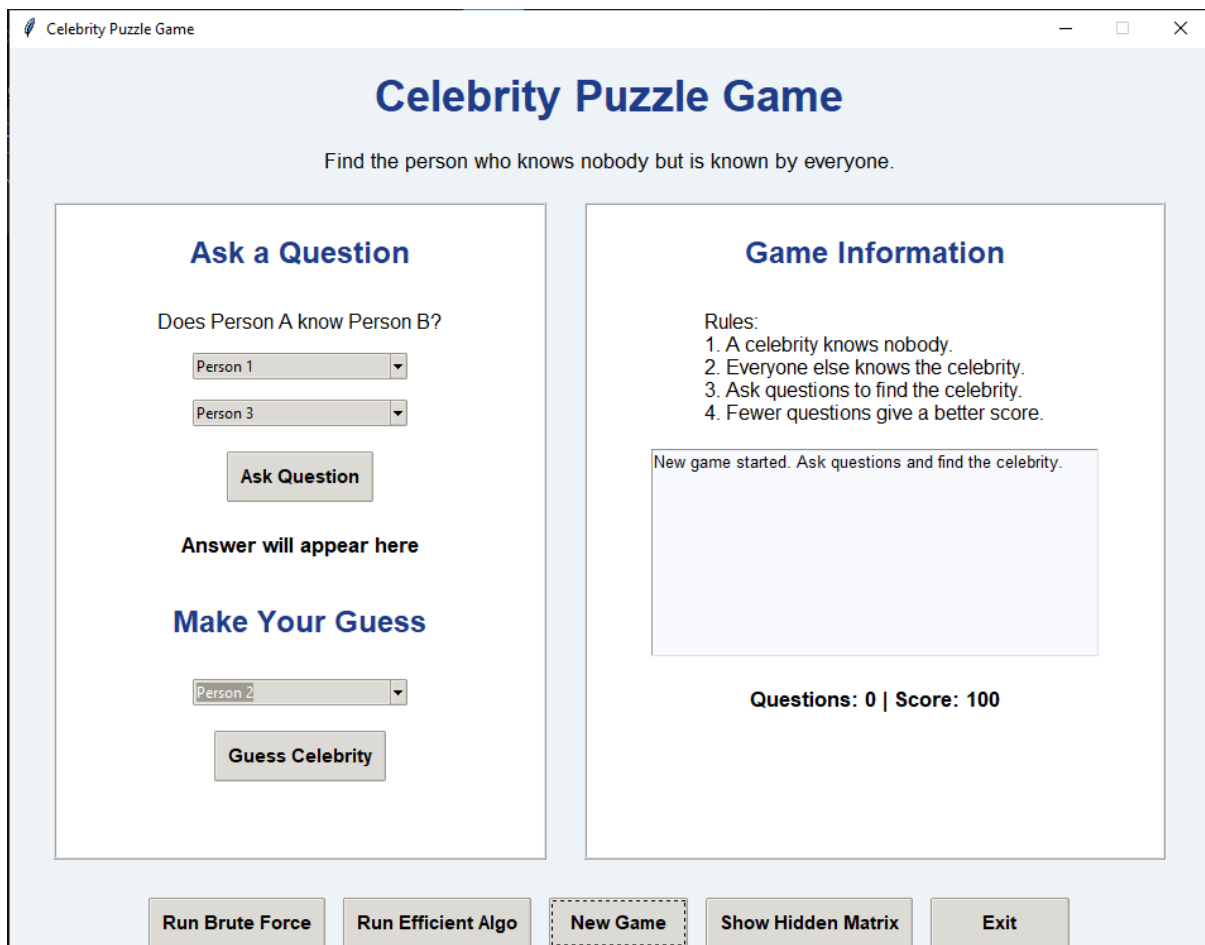
The .py code file is attached with the report.

[Open .py file](#)

GitHub Link:

<https://github.com/Adeel-Yaqoob/DAA-project>

6. Sample Output Screenshots:



The screenshot shows the main interface of the 'Celebrity Puzzle Game'. The window title is 'Celebrity Puzzle Game'. The main heading is 'Celebrity Puzzle Game' in a large, bold, blue font. Below it, the instruction 'Find the person who knows nobody but is known by everyone.' is displayed. The interface is divided into two main panels. The left panel, titled 'Ask a Question', contains a question 'Does Person A know Person B?' with two dropdown menus for 'Person 1' and 'Person 3'. Below these is an 'Ask Question' button. Underneath, it says 'Answer will appear here'. Further down, the section 'Make Your Guess' has a dropdown menu for 'Person 2' and a 'Guess Celebrity' button. The right panel, titled 'Game Information', lists the rules: 1. A celebrity knows nobody. 2. Everyone else knows the celebrity. 3. Ask questions to find the celebrity. 4. Fewer questions give a better score. Below the rules, a text box says 'New game started. Ask questions and find the celebrity.' At the bottom of this panel, it shows 'Questions: 0 | Score: 100'. At the very bottom of the window, there is a row of five buttons: 'Run Brute Force', 'Run Efficient Algo', 'New Game' (which is highlighted with a dashed border), 'Show Hidden Matrix', and 'Exit'.

Figure 1: Main Interface of Celebrity Puzzle Game

Ask a Question

Does Person A know Person B?

Person 1

Person 3

Ask Question

NO, Person 1 does not know Person 3.

Figure 2: Asking A Question

Make Your Guess

Person 1

Guess Celebrity

Wrong Guess

 Wrong guess! The celebrity was Person 4.

OK

Figure 3: Guessing the Celebrity

Hidden Matrix

Hidden Knows Matrix

	P1	P2	P3	P4	P5
P1	0	1	1	0	1
P2	0	0	1	1	0
P3	0	0	0	0	0
P4	0	1	1	0	0
P5	0	1	1	1	0

Celebrity: Person 3
1 = knows, 0 = does not know

Figure 4: Showing Hidden Matrix

7. Conclusion:

This project successfully implemented an interactive Celebrity Puzzle Game using Python and Tkinter GUI. The game showed the practical implementation of the Celebrity Problem and allowed the users to understand algorithmic problem solving interactively.

Brute force and Decrease-and-Conquer both algorithms were implemented and compared. The efficient algorithm performed better with linear time complexity of the brute force approach.

The graphical user interface made the game more user-friendly, interactive and suitable for showing algorithm concepts visually.
